

1. The problem in plain English

Audience: Charles, re-entering after time away

You do not need any other document open while reading this.

Binding insight (read early): The hero shows **outcomes** (draft rate vs pool quality). **Environment** = $L_{net} = B - D$. **Advancement** = score (S_i, λ) then **select** (top K now) — separate from describing the peer environment. See doc 02 and `../BINDING_Selection_is_its_own_step.md`.

The one-sentence dissertation idea

Organizations advance some people and not others. Being surrounded by **strong peers** can **help** you (you learn, you look good by association) and **hurt** you (you are harder to notice, you compete for scarce slots). The dissertation asks whether that **tension** shows up in data — and whether a **simple mechanism** (crowding in the score that ranks who advances) can produce curves that look like the data.

Basketball is one **test bed**: college players who get drafted to the NBA.

The basketball “hero” — what we already see in the data

Think of one row per **player-season**: a college player in a given year, on a given team.

Question: Does the chance of eventually being **NBA drafted** relate to the **quality of his teammates**?

We measure teammate quality using box-score performance (points per minute, adjusted within season). Crucially, we use **leave-one-out** quality: average teammate performance **excluding the player himself**, so he is not counted in his own pool average.

Call that number **pool quality (leave-one-out)**. Higher means better teammates around him.

What we plot: Split all player-seasons into 16 equal-sized bins from lowest to highest pool quality. For each bin, compute the **fraction who were ever drafted**. Connect the dots.

What we see (qualitative shape):

- Draft rate **rises** as pool quality increases through the middle bins.
- Rate **levels off** in high-but-not-top bins.
- In the **very top bin only**, rate **drops** — an inverted-U **tail**, not necessarily a symmetric hill.

That plot is the **hero**. It is a **stylized fact** — something real in the NCAA panel under fixed rules (which seasons, which minutes filter, which binning). It is **not** yet a story about why NBA teams behave that way.

First step — two axes, two curves (real data): Before the hero is “surprising,” check the obvious axis. Bin the **same** player-seasons by **own college performance** (same ppm stat, z-scored within season) instead of teammate pool quality. Draft rate then rises **monotonically** from the bottom talent bin to the top (~0% → ~7% on the locked July 2026 sample). That is the **talent baseline** — talent matters, as it should. The **hero** bins on **pool quality** instead and gets a **different** shape: rise, plateau, then a **dip in the top pool-quality bin only**. The phenomenon is the **second** curve, not draft itself. We are **not** claiming the NBA ignores talent; we are showing that **peer environment** has its own non-monotone pattern once you look at it on its own axis.

Important: The hero shows **who got drafted** vs pool quality. It does **not** tell you whether the bend comes from peer effects on **development** or from **congestion in the ranking/score** that feeds who gets selected. That split is Layer B (doc 02): environment ($B - D$) vs **score then select**.

Sample size (locked July 2026): about 62,000 player-seasons; about 1,100 drafted.

Why this matters for the “simplified model” push

Alex’s scientific arc (paraphrased): show the **phenomenon**, then show a **minimal generative rule** where **talent alone is not enough** — you need something like **congestion in the score** — then later add richer measurements and predictions.

You are **not** being asked, in the next few days, to:

- perfectly reproduce every bin of the hero from simulation;
- estimate separate “benefit” and “congestion” functions from one chart;
- unify Army, basketball, and tenure in one model.

You **are** being asked to hold a clear line:

1. **Here are the two empirical curves** — talent baseline (monotone) and hero (pool quality with tail dip); both on real NCAA data under the locked spec.
2. **Here is a disciplined artificial league** where **scoring** purely on talent gives a different curve than scoring on talent **minus** a congestion penalty — same winner rule (top K), side by side, honest axes, explicit limits on what we claim.

That is the simplified model **deliverable** for basketball v1. Slide summary: `Model.pdf`.

Words you will see again (defined once here)

Term	Plain meaning
Panel	The main analysis table of real college player-seasons — one row per athlete × season × team, with own stats, leave-one-out teammate quality, and draft outcome; we filter it, then bin it for the hero
Hero	The binned draft-rate plot vs leave-one-out pool quality
Talent baseline	Same panel binned by own performance instead of pool quality; draft rate rises monotonically (sanity check before the hero)
Pool quality (LOO)	How good your teammates are on paper, with you removed from the average
Draft rate	Share of players in a bin who were ever NBA drafted
Stylized fact	A robust pattern in data under stated rules — not yet causal proof
Generative model / sim	A computer-made fake league with explicit draft rules
Congestion	“Too many good peers” — harder to stand out or win a scarce slot
BINDING	Charles-locked rule: keep L_net (environment) separate from advancement; keep score (S_i) separate from select (top K / later noise)

Emotional checkpoint

If the only thing you take from this page is:

Good players get drafted more than weak ones (talent baseline). But among teammate environments, the very best pools do not have the highest draft odds — and we have a fixed hero plot that shows that.

—you have enough to read document **02** (three kinds of model) without guilt about everything else in the repo.

Next: [02_Three_Kinds_of_Model.md](#)